

Corporate Travel Dispatch Intelligence (CTDI)

Executive Summary — Version 1.0.0 (First Public Release)

Developed and operated by CS Executive Services, LLC | Source: github.com/CorporateTravelDC | Released June 29, 2026

Overview

CTDI is an autonomous dispatch intelligence platform for executive transportation operators. It continuously monitors commercial aviation, rail, weather, and airspace-restriction data across 14+ live feeds — including a dedicated ACARS/VDL2 avionics datalink layer decoded directly from the VHF band — distills those signals into actionable alerts and a single operational go/no-go score, and pushes them to dispatcher devices within seconds of a change. The platform is designed for any operator to deploy on hardware they already own: the complete stack runs on a Raspberry Pi 5 or any Fedora/Linux host (aarch64 or x86_64), with no cloud dependency of any kind.

CTDI is built for operators who answer to regulators and to regulated clients. Its decision engine is designed around FAA Part 135.609 go/no-go criteria, its airspace monitoring is aligned to FAA NOTAM frameworks for the FRZ, SFRA, P-56, and TFRs, and its local-first architecture keeps all passenger and movement data on-premises — a compliance posture suited to government, legal, financial, and executive-protection clientele.

Version 1.0.0, published June 29, 2026 under the CorporateTravelDC GitHub organization, is the first public release. The Washington, DC configuration ships as the reference implementation, and a documented regionalization path moves the platform to any metro area — including European and Asia-Pacific feed equivalents.

Live Data Feeds

CTDI operates a hybrid pull/push feed architecture with a third, RF-based layer on top: REST feeds are polled on fixed intervals (five minutes for most sources); FAA SWIM and NWS push feeds deliver events in real time when connected; and ACARS/VDL2 aviation datalink messages are decoded continuously from the VHF band by dedicated on-site receivers. A heartbeat-based failover hands each feed back to REST polling automatically within 90 seconds if a push connection drops — no manual intervention required.

Feed	Content	Source type	Cadence
ACARS	Aircraft datalink — clearances, ATIS, weather uplinks, position and maintenance messages	RF avionics datalink	Real time
VDL2	VHF Data Link Mode 2 aircraft operational messages	RF avionics datalink	Real time
METAR	Surface weather observations	REST	5 min
NWS Alerts	Hazardous weather warnings	REST	5 min
TFR	Temporary flight restrictions	REST + push	5 min / live
NOTAMs	FAA notices to air missions	REST	5 min
Amtrak	Rail status and delays	Push + poll	Live / 5 min

Feed	Content	Source type	Cadence
ATCSCC Ops Plan	Daily national airspace plan	REST	1 hr
NAS Programs	Ground delay programs, ground stops	REST	5 min
FDPS	Flight plans and track data	FAA SWIM push	Real time
STDDS	Surface and terminal tracks, TFRs	FAA SWIM push	Real time
TFMS	Traffic management initiatives	FAA SWIM push	Real time
AIM / FNS	Digital NOTAMS	FAA SWIM push	Real time
TBFM	Arrival sequencing	FAA SWIM push	Real time
ITWS	Terminal-area weather	FAA SWIM push	Real time
NWWS-OI	NWS products via XMPP push	Push	Real time
ADS-B	Live aircraft within 250 NM	REST proxy	30 sec cache

FAA SWIM and NWWS-OI push feeds require operator-provisioned credentials; the platform activates them automatically once credentials are entered — no code changes required. The ACARS/VDL2 layer requires only an antenna and SDR receiver on-site.

Avionics Source of Truth: ACARS & VDL2 Integration

Most travel-intelligence products see aircraft the way the public does: radar-derived ADS-B position, often republished secondhand by web trackers. CTDI adds a fundamentally different data layer — it receives actual flight operational data from aircraft avionics, decoded on-site from FAA-regulated aviation datalink infrastructure. The platform hears what the aircraft hears.

Three dedicated containers form the avionics datalink chain:

Container	Role
corporatetravelcdc-dumpvdl2	VHF Data Link Mode 2 (VDL2) decoder — captures aircraft operational messages directly from the VHF band.
corporatetravelcdc-acarshub	ACARS message hub — decodes Aircraft Communications Addressing and Reporting System transmissions: departure clearances, ATIS, weather uplinks, position reports, and maintenance messages.
corporatetravelcdc-acarsrouter	Routes, deduplicates, and normalizes ACARS / VDL2 message streams into the dispatch intelligence layer.

Why it matters for schedule integrity. ACARS carries a class of signal that is structurally invisible to ADS-B-only systems: pre-departure clearance activity, taxi state, airborne weather and ATIS uplinks, position reports, and aircraft-health messages. A departure clearance on the datalink means wheels-up is coming — before the aircraft has moved a foot. A maintenance message can foreshadow a delay or equipment swap before any schedule system reflects it. For a dispatch operation whose margins are measured in minutes at the curb, this is the difference between reacting to a change and anticipating it.

- **Pre-departure clearances (PDC):** departure activity visible before movement — the earliest reliable wheels-up predictor.
- **OOOI and taxi state from the airframe:** out, off, on, in — reported by the aircraft itself, not inferred from position.
- **Weather and ATIS uplinks:** what the crew is actually being told about conditions at the destination.
- **Maintenance and health messages:** early warning of delays, gate returns, and equipment substitutions.

The router container deduplicates and normalizes both message streams into the same SQLite intelligence layer that powers CPS scoring, watchlist alerting, and the pattern archive — so avionics-derived signals participate in every downstream decision the platform makes.

Critical Predictability State (CPS)

CPS is CTDI's go/no-go scoring engine, explicitly designed around the go/no-go decision criteria of FAA Part 135.609 — the pre-flight risk assessment regulation written for on-demand air charter operations — and applied continuously to ground operations. Six live factors are evaluated and combined: ceiling, visibility, wind, and precipitation (from METAR data), active TFRs and restricted airspace, and FAA ground delay programs. The output is a single unambiguous state:

- **GREEN / GO** — conditions support normal operations.
- **YELLOW / MARGINAL** — one or more factors degraded; heightened monitoring advised.
- **RED / NO-GO** — conditions outside safe operating limits.

The score recomputes every 60 minutes and on demand, and every state change is pushed to subscribed devices. Dispatch decisions become scored, logged, and defensible rather than ad hoc — a posture that matters when clients or auditors ask why a run was held or re-routed.

Priority Alerting — Marine One / AF1 Hot-Push

VIP and presidential movements activate TFRs that reshape both airspace and surface routes around major airports with minimal notice. CTDI polls the TFR feed on a five-minute cadence (supplemented by real-time SWIM push when connected) and, on detecting a VIP/POTUS TFR signature, fires a maximum-priority (P5) ntfy push that breaks through silent and do-not-disturb modes. This alerting operates against the FAA NOTAM and Secret Service airspace-security frameworks that govern national-capital-region movements.

- **Dual-topic delivery:** full detail to the domain topic (e.g., tfr-alert, flight-alerts), a concise bottom line to the dispatch topic.
- **Broad alert coverage:** OOOI flight events, diversions, rail delays, hazardous weather, CPS changes, feed-health audits, and daily operational briefs — each on its own subscribable channel.
- **Alert hygiene:** a five-minute deduplication window suppresses re-fires of the same event, preventing alert fatigue.

Pattern Intelligence & Anomaly Detection

A built-in archive recorder captures rolling snapshots of every live feed into a local database — every five minutes, 24/7, zlib-compressed so a full 52-week archive fits under 500 MB. Over successive quarters, this longitudinal dataset surfaces patterns invisible in real-time views:

- Recurring VIP TFR activation windows at key airports — the basis for standing client advisories.
- Ground delay program frequency by airport and season — historically predicts peak-quarter hub congestion.
- Rail on-time performance drift across corridors — grounds mode recommendations in data rather than anecdote.
- Airport construction and closure windows from NOTAM history — by corridor and month.

The same archive powers a zero-credential demo mode: historical snapshots replay through the live REST API surface, producing a fully real-looking dashboard for evaluations with no live feeds or credentials required. Retention tiers (2 to 52 weeks) are tracked automatically for quarterly and annual reporting readiness.

Architecture & Privacy

Local-first by design

Eight rootless Podman containers are managed by systemd Quadlets. Five core services — web (FastAPI REST API), poller (feed scheduler and AI skills), pusher (alert dispatcher), ingest (SWIM/NWS push), and runner (React PWA operator dashboard) — share a single SQLite database in WAL mode, joined by the three-container avionics datalink chain (dumpvdl2, acarshub, acarsrouter) that feeds the same intelligence layer. There is no external database server, no message broker, and no cloud control plane.

Privacy and data sovereignty

- **Zero data egress:** no itinerary, passenger, or movement data leaves the device. All AI inference (daily briefs, TFR enrichment, operator chat) runs on-host via Ollama with local models — no cloud LLM key exists in the system.
- **Tiered access control:** anonymous read-only data, Tailscale-gated operational endpoints, and hashed bearer tokens for administration. Plaintext tokens are displayed once at creation and never stored.
- **Audit-ready:** append-only audit log with 90-day retention that never leaves the host; sensitive radio data is excluded from every release artifact by policy.

Operator dashboard

The runner container serves a WCAG AA-compliant progressive web app: live map with airspace rings and TFR polygons, global ADS-B aircraft overlay with watchlist highlighting, rail map, live feed panel, alert stream, and a chat drawer backed by the local LLM. Installable to any phone, tablet, or desktop as a PWA.

Regulated Industry Deployment

CTDI operates at the intersection of three regulated domains: executive ground transportation (DOT/FTA), FAA-regulated airspace (FRZ, SFRA, TFRs, P-56), and FAA-regulated aviation communication infrastructure (ACARS/VDL2 datalinks). That positioning is deliberate, and it shapes both the engineering and the deployment posture:

- **Regulatory-grade decisioning:** the CPS engine is designed around FAA Part 135.609 go/no-go criteria for on-demand air charter — dispatch decisions inherit a regulatory framework rather than an in-house heuristic.

- **Regulated data infrastructure:** the avionics layer is built on FAA-regulated communication links, decoded on-site — not scraped web data of uncertain provenance or terms of use.
- **Airspace-security awareness:** Marine One / AF1 TFR alerting touches the Secret Service and FAA NOTAM compliance frameworks that govern national-capital-region operations.
- **Data sovereignty by architecture:** local-first design means no PII or client movement data transits third-party infrastructure — a material compliance differentiator for finance, law, government contracting, and healthcare executive transport.
- **Discretion as a requirement, not a courtesy:** privacy commitments align with the NDAs, security protocols, and compliance obligations under which government, legal, financial, and executive-protection clients operate.
- **Cleared and sensitive environments:** suitable for operators serving cleared personnel, government contractors, and CJIS-adjacent environments where cloud-hosted movement data is a non-starter.

Platform Availability

Platform	Status	Notes
Fedora 43+ (x86_64 / ARM64)	Verified	Fully tested reference platform
Raspberry Pi OS (Pi 5, aarch64)	Verified	Fully tested reference hardware
macOS (Apple Silicon / Intel)	Community	Podman Machine or Docker Desktop
Windows x64	Community	WSL2 installer; Ollama runs natively
Android ARM64 (Termux)	Community	Bare Python; lightweight local models
iOS / iPadOS	Web client	PWA via browser; Add to Home Screen

Reference deployment target: Raspberry Pi 5 (BCM2712, aarch64). The identical codebase scales to workstation- and server-class Linux hosts.

Deployment & Integration

A new deployment moves from git clone to live dispatch in an afternoon: clone the repository, run setup.sh (creates the system user, directories, configuration from templates, Quadlets, and nginx vhosts), build the container images locally, and start the services. Feeds begin polling immediately; the avionics datalink chain comes up with them wherever an SDR receiver is attached.

- **Reservation-system integration:** outbound webhooks from the operator's existing booking platform automatically add flights and trains to the CTDI watchlist — OOOI phase tracking, delay and diversion alerts begin with zero manual entry.
- **Regionalization:** three documented configuration points (hub airport list, weather alert area, forecast-office filter) move the platform to any metro area; European and Asia-Pacific feed equivalents are documented.
- **White-label deployment:** any executive transportation operator can deploy CTDI under their own brand and operating area.
- **Release integrity:** all public releases are GPG-signed; active signing keys are published in-repo by full fingerprint.

Contact

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